

Chapter

2

DC Electrics - Switches

Switches	29
Proximity Detectors.	30
Time Switches	33
Centrifugal Switches	33

Switches

The initiation and control of aircraft circuits is achieved by switches and relays. Some typical switches are described here.

Toggle Switch

A general purpose switch common in older aircraft having a number of isolating contacts inside. It can be a two position switch (on or off) or a multi-position switch sprung biased to the centre or off position and then pressed and held to select in the desired direction.

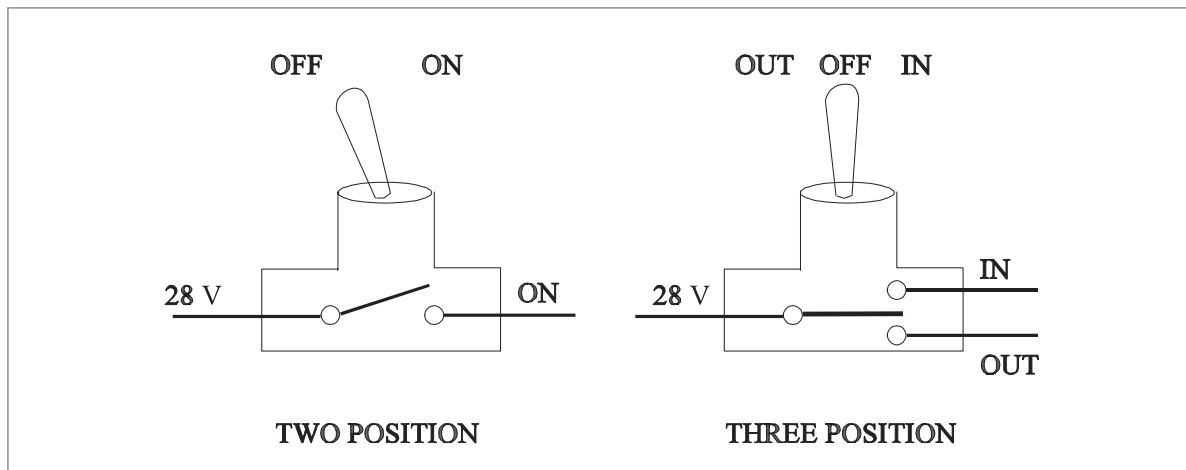


Figure 2.1

Switch Light

Switch lights have largely replaced toggle switches in modern aircraft and combine the functions of a switch with a push action and an indicator light for the associated function.

There are two basic types

- **Momentary action** press and hold to activate, release to deactivate.
- **Alternate action** press and release to activate, press and release a second time to deactivate.

The indicator in the lens confirms the selected position or provides a warning which requires the switch to be selected.

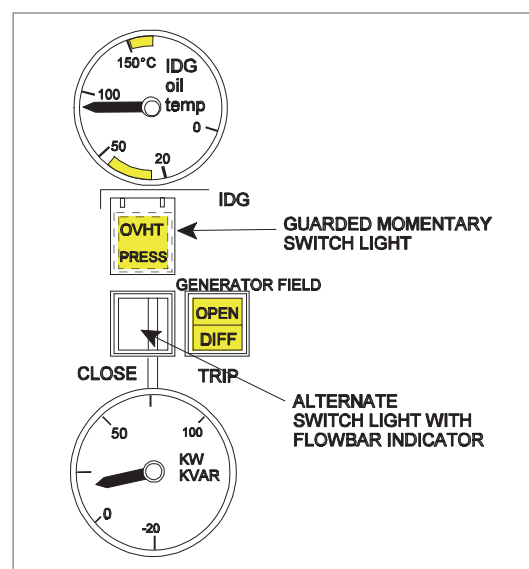


Figure 2.2

Guarded Switches

Toggle switches or switch lights can be guarded to prevent inadvertent operation, e.g. generator disconnects the fuel dump master. (See previous diagram)

Microswitch

Microswitches are still used in modern aircraft to detect the position of a particular device e.g. door opened or closed.

The name Microswitch describes the small movement between the 'make and break' position. Microswitches can activate indications on the flight deck or control relays for a sequenced operation. They are largely replaced by proximity detectors on modern aircraft.

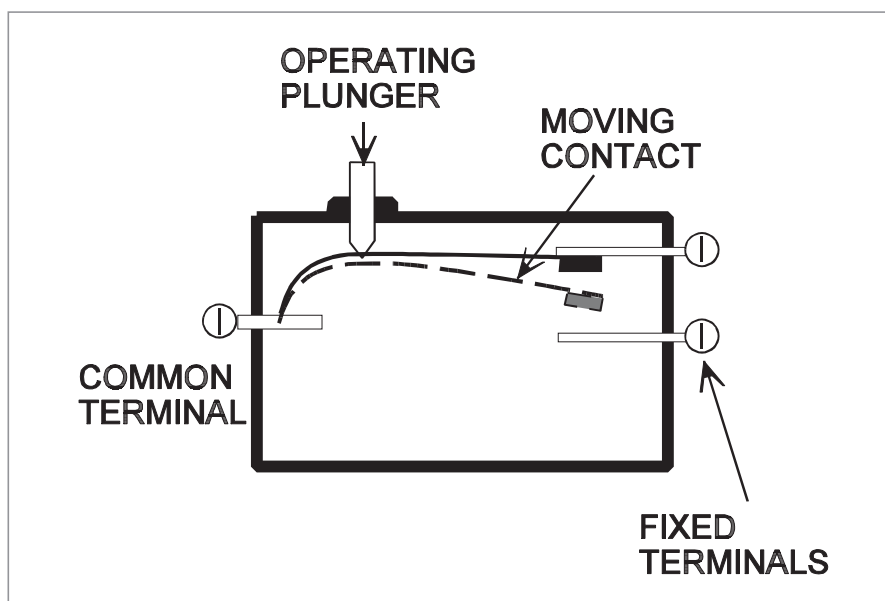


Figure 2.3 Microswitch

Bimetallic Switch (Thermal Switch)

Bimetallic switches are temperature sensitive switches and are activated when a certain value of temperature is reached to provide an indication to the pilot or to activate / deactivate a circuit, e.g. fire detection circuits, battery overheat switch, oil temperature warning light.

Proximity Detectors

Proximity detectors are electrical or electronic sensors that respond to the presence of a material. The electrical or electronic response is used to activate a switch, relay or transistor. There are many types of proximity detectors, the major types being inductive, capacitive and magnetic. The inductive and magnetic sensors need the monitored material to be metal, but the capacitive type can monitor either metal or non-metal materials.

Inductive Type

This type of sensor has an inductance coil whose inductance changes when a ferromagnetic material (target) is brought into close proximity with it.

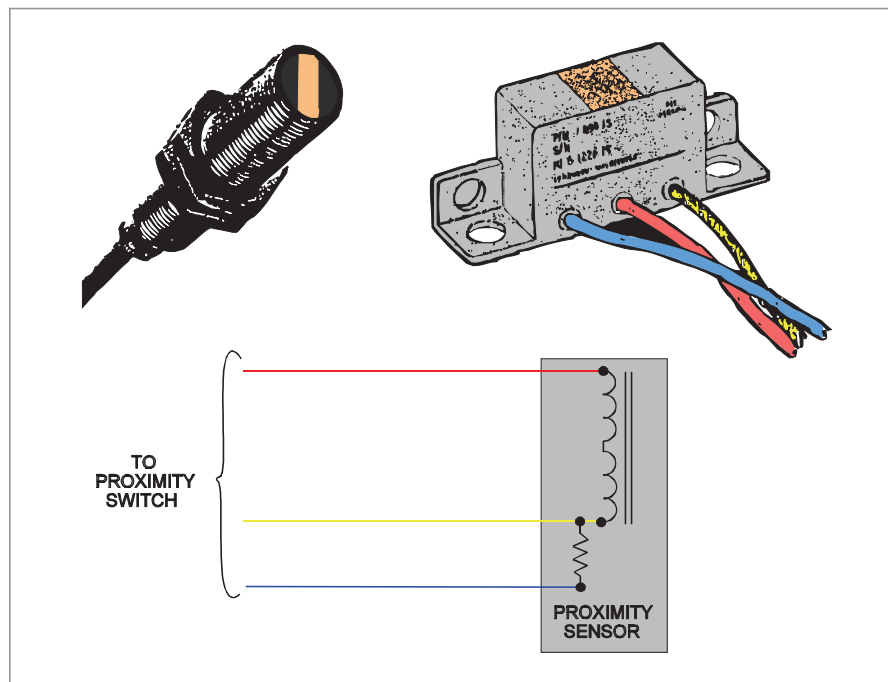


Figure 2.4

This type of sensor is used in undercarriage systems in place of microswitches. A typical undercarriage system is described below. Each proximity switch consists of three components:

- A printed circuit card located in what is called the landing gear accessory unit.
- A sensor located on appropriate landing gear structure.
- An actuator (or target) for each sensor, located adjacent to its sensor.

The proximity sensor is a hermetically sealed unit, and is actuated by the presence of the actuator or target, i.e. it is not touched by it. As a result, the proximity switch is unaffected by atmospheric conditions, and is highly reliable.

Capacitive Type

In this type of sensor detection is made by a capacitor undergoing a capacitance change owing to the proximity of material.

The capacitive proximity detector is an extremely versatile device in that it is capable of detecting all materials, liquid and solid. As well as detecting the presence of a ferrous or non-ferrous target, it can be used to detect high or low liquid levels in a hydraulic or fuel system.

Magnetic Type

A coil situated in a magnetic field will have an electromotive force (EMF) induced in it if the magnetic flux changes. The magnitude of the induced EMF will depend on the rate at which the flux is changed. These are the basic principles on which the magnetic proximity detectors operate.

In its simplest form, a coil is wound around a bar magnet and one pole of the magnet is then located close to a ferrous object. If the ferrous object moves, the flux in the magnet changes and an EMF is induced in the coil. If a number of ferrous objects move past the magnet, a train of pulses is induced in the coil.

Magnetic detectors are most commonly used in conjunction with mild steel gear wheels, each tooth in the wheel being, in effect, a ferrous object. The detector is located radially and close to the periphery of the wheel and provides an output having a frequency equal to the frequency of passage of the teeth past the detector.

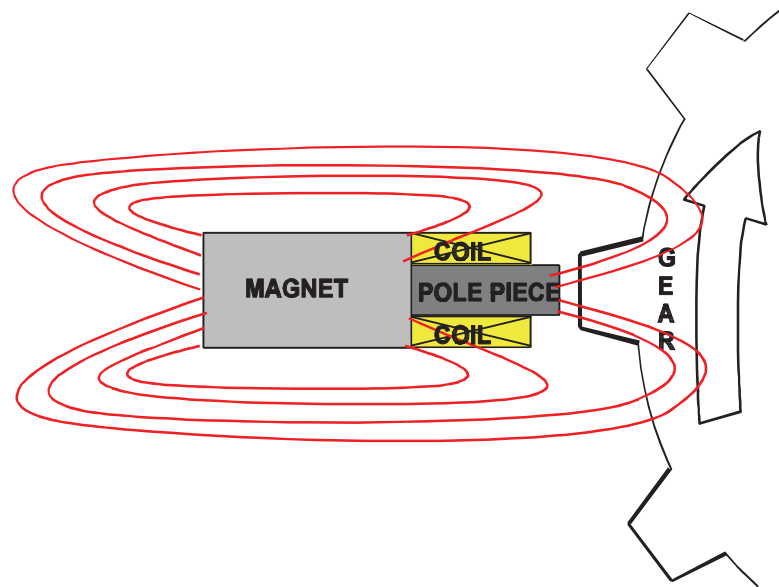


Figure 2.5

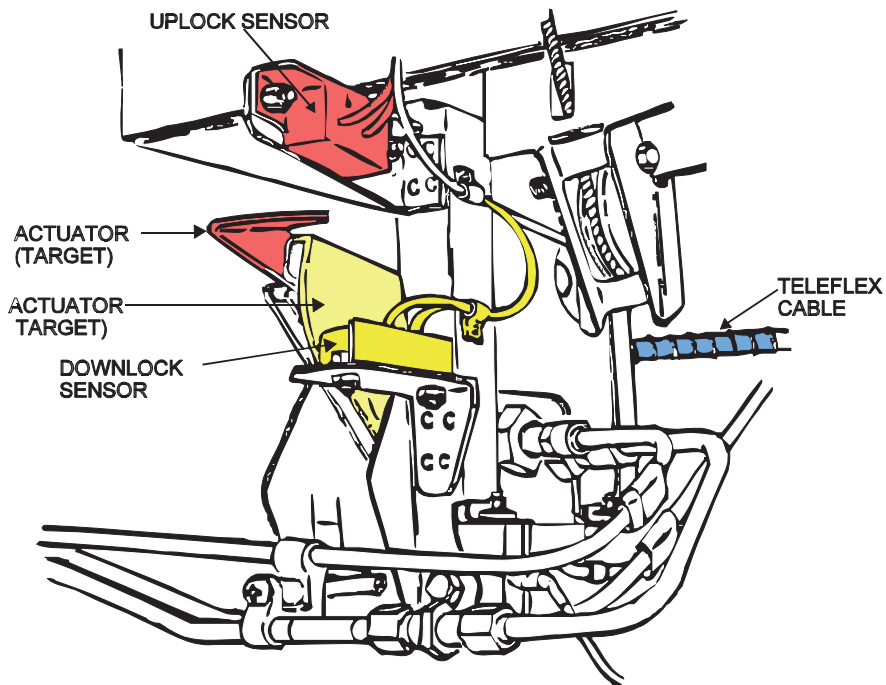


Figure 2.6

Time Switches

Time switches or relays can be initiated electrically or mechanically to activate a circuit after a specific time interval has occurred, e.g. auxiliary power unit air intake door closes 30 seconds after APU has shut down.

Centrifugal Switches

These can be set to activate or de-activate a circuit as the RPM of a device increases or decreases, e.g. starter motor cut-out switch.

Chapter

3

DC Electrics - Circuit Protection and Capacitors

Electrical Faults	37
Circuit Protection Devices.	37
Fuses	37
The Cartridge Fuse	37
Spare Fuses	38
High Rupture Capacity (HRC) Fuses	39
Dummy Fuses	39
Current Limiters	39
Circuit Breakers	40
Reverse Current Circuit Breakers	41
Capacitors	41
Capacitance	42
Capacitor in a DC Circuit	42
Capacitor in an AC Circuit	43
Capacitors in Parallel	44
Capacitors in Series	44
Questions - Circuit Breakers	45
Questions - Fuses	47
Answers - Circuit Breakers	50
Answers - Fuses	50

Electrical Faults

In an electrical circuit, abnormal conditions may arise for a variety of reasons, which can cause overcurrent or overvoltage conditions.

If allowed to persist, these abnormal conditions or faults will lead to damage or destruction of equipment and in extreme cases, loss of life. Certainly the essential power supplies will fail, and it is therefore necessary to protect circuits against all such faults, by the use of fuses and circuit breakers.

Circuit Protection Devices

There are a number of protection devices used in aircraft electrical systems but only 2 basic types are discussed here:

- Fuses
- Circuit breakers

The fundamental difference in the type of protection provided by fuses and circuit breakers is in their time of operation relative to the attainment of maximum fault current.

A fuse normally opens the circuit before full fault current is reached, whereas the circuit breaker opens after the full fault current is reached.

This means that when circuit breakers are used as the protection device, both the circuit breaker and the component must be capable of withstanding the full fault current for a short time.

The circuit breaker has the capability, which the fuse has not, of opening and closing the circuit, and can perform many such operations before replacement is necessary. It may also be used as a circuit isolation switch.

Fuses

There are 3 basic types of fuse currently in use on aircraft:

- Cartridge fuse
- High rupture capacity (HRC) fuse
- Current limiter fuse

The Cartridge Fuse

The cartridge type fuse consists of a tubular glass or ceramic body, 2 brass end caps and a fuse element.

The element may be one of the following:

- Tinned copper wire
- Silver wire
- A strip of pure zinc - electro tinned

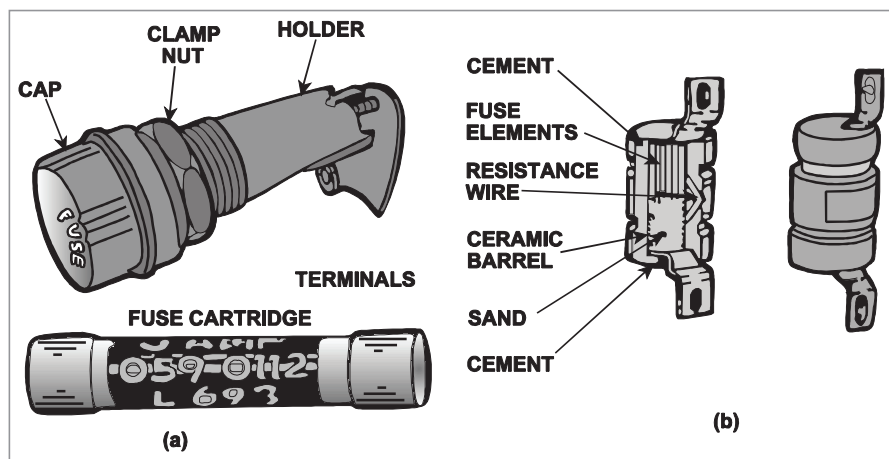


Figure 3.1 Typical fuses

a. A light duty circuit fuse

b. A high rupturing capacity fuse

The latter type element is generally used in heavy duty circuits, the zinc strip being cut to a specified width.

A fuse operates when the current flowing through it is sufficient to melt the wire or strip element, the time taken varying inversely with the current.

All fuses are rated at a specific current value, i.e. the rating indicates the current they will carry continuously or intermittently without unduly heating up or deteriorating.

The rating of a fuse for a particular circuit is such that it is not less than the normal current flowing in the circuit, but that it operates ('blows') at a current level below the safety limit of the equipment or cable used.

For this reason only the specified fuse should be used in a particular circuit. The diagram shows typical aircraft fuses; the ratings can vary between 0.5 and 500 amps, the higher ratings being limited to the HRC or current limiter types.

Fuses are made of a type of wire which has a low melting point, and when it is placed in series with the electrical load it will melt, blow or rupture when a current of higher value than its ampere rating is placed upon it.

Fuses are rated in 'amps'.

A blown fuse may be replaced with another of the correct rating once only. If it blows again when switching on, there is a defect in the system and the fuse must not be changed again until the circuit has been investigated.

Spare Fuses

The carriage of spare fuses is mandatory, the quantity of spares being at least 10% of the number of each rating installed, with a minimum of 3 of each.

High Rupture Capacity (HRC) Fuses

The high rupture capacity (HRC) fuse is an improvement on the cartridge type fuse. It is used mainly for high current rated circuits.

The body is a ceramic material of robust construction and has one or more element holes. The element holes are filled with powdered marble or clean quartz sand. The end caps are of plated brass or copper.

The HRC has the following advantages over the normal glass cartridge type:

- more accurate operation
- operates without flame
- does not deteriorate with age
- is more robust
- operates rapidly
- is not affected by ambient temperature

Dummy Fuses

Aircraft electrical circuits which are not in use will have dummy fuses fitted. If it is necessary to isolate a particular circuit by the removal of the fuse in order that the system be made 'safe' or for work to be carried out, a dummy fuse or fuse holder should replace the fuse which has been removed.

To distinguish the dummy fuse, a red streamer is attached to it.

Dummy fuse links are manufactured to standard fuse dimensions from red plastic, the centre portion being square in section with corrugated sides to facilitate identification.

Services protected by circuit breakers are made safe in a similar manner, a warning flag or plate is clipped to the tripped circuit breaker, indicating that the service has been rendered safe for servicing.

Current Limiters

Current limiters, as the name suggests, are designed to limit the current to some predetermined amperage value.

They are also thermal devices, but unlike ordinary fuses they have a **high melting point**, so that their time/current characteristics permit them to carry a considerable overload current before rupturing.

For this reason their application is confined to the protection of heavy-duty power distribution circuits. The output of a Transformer Rectifier Unit would be a prime location for a current limiter to be used.

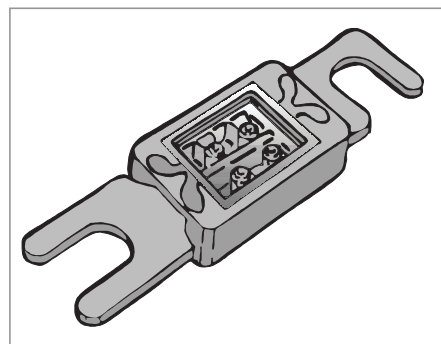


Figure 3.2 A typical current limiter (an airfuse)

A typical current limiter (manufactured under the name of 'Airfuse') is illustrated in Figure 3.2, it incorporates a fusible element which is, in effect, a single strip of tinned copper, drilled and shaped at each end to form lug type connections, with the central portion 'waisted' to the required width to form the fusing area.

The central portion is enclosed by a rectangular ceramic housing, one side of which is furnished with an inspection window which, depending on the type, may be glass or mica.

Circuit Breakers

Circuit breakers combine the function of fuse and switch and can be used for switching circuits on and off in certain circumstances.

They are fitted to protect equipment from damage resulting from overload, or fault conditions. The design and construction of CBs is wide and varied.

Generally, the CB incorporates an automatic thermo-sensitive tripping device and a manually or electrically operated switch.

Some electrically operated CBs may also include electromagnetic and reverse current tripping devices.

The smaller type single button CBs, shown in [Figure 3.3](#), range from 5 amps to 45 amps, whereas the larger reverse current CBs can be rated up to 600 amps.

The diagram shows two typical CBs, the single push pull button type has a white marker band to assist in identifying a 'tripped' circuit breaker amongst a panel of many.

The CB at (b) is fitted with a "manual trip" button and is more usually associated with a heavy duty circuit.

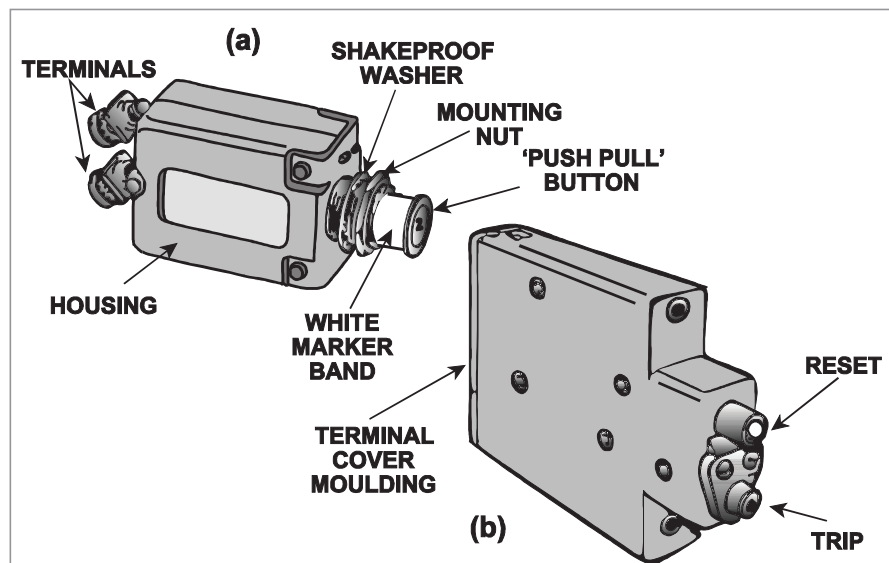


Figure 3.3 Circuit breakers

CBs are common on the flight deck of modern aircraft and can be categorized as either:

- a Non-trip Free Circuit Breaker, or
- a Trip Free Circuit Breaker.

The **non-trip free circuit breaker** may be held in under fault conditions and the circuit will be made, this is clearly dangerous.

The **trip free** circuit breaker if held in under the same circumstances, the circuit can **not** be made.

Pressing the re-set button will reset either CB if the fault has been cleared.

Reverse Current Circuit Breakers

These CBs are designed to protect power supply systems and associated circuits against fault currents reversing against the normal current direction of flow of a magnitude greater than those at which cut-outs normally operate.

They are furthermore designed to remain in a "locked-out" condition to ensure complete isolation of a circuit until a fault has been cleared.

Capacitors

Introduction:

A capacitor can perform three basic functions:

- Stores an electrical charge by creating an electrical field between the plates.
- Will act as if it passes Alternating Current
- Blocks Direct Current flow

Construction:

In its simplest form a capacitor consists of two metal plates separated by an insulator called a **dielectric**. Wires connected to the plates allow the capacitor to be connected into the circuit.

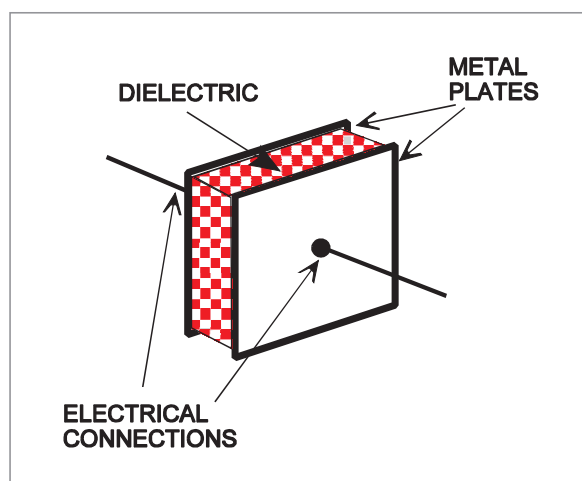


Figure 3.4 The construction of a simple capacitor

Symbols:

Figure 3.5 shows the electrical circuit symbols for various capacitors. With the polarized capacitor it is important to connect the positive terminal to the positive supply. Non-polarized types can be connected either way round.



Figure 3.5 Capacitor symbols

Capacitance

The capacitance (C) of a capacitor measures its ability to store an electrical charge. The unit of capacitance is the **FARAD** (F). The farad is subdivided into smaller, more convenient units.

1 microfarad (1 μF)	= 1 millionth of a farad	= 10^{-6} F
1 nanofarad (1 nF)	= 1 thousand millionth of a farad	= 10^{-9} F
1 picofarad (1 pF)	= 1 millionth millionth of a farad	= 10^{-12} F

Factors Affecting Capacitance:

- Area of the plates - a large area gives a large capacitance
- Distance between the plates - a small distance gives a large capacitance
- Material of the dielectric - different materials have different values of capacitance, for example paper, mica, air and fuel. The value of the dielectric is referred to as the **dielectric constant (k)**. For example, waxed paper has a k value of about 3, whereas air has a k of 1. So a capacitor having waxed paper as its dielectric would have 3 times the capacitance of the same capacitor having air as its dielectric.

Working Voltage:

This is the largest voltage DC or Peak AC which can be applied across the capacitor. It is often marked on the case of the capacitor and if it is exceeded, the dielectric may break down and permanent damage result.

Capacitor in a DC Circuit

Figure 3.6 shows a capacitor in series with a battery and a switch. If the switch is closed, electrons are pushed by the battery on to plate Y building up a negative charge. This charge exerts a repelling force across the dielectric which causes electrons to leave the plate X and be attracted to the positive plate of the battery. While this charging action is taking place electrons are passing through the connecting wires **but no current flows through the dielectric**.

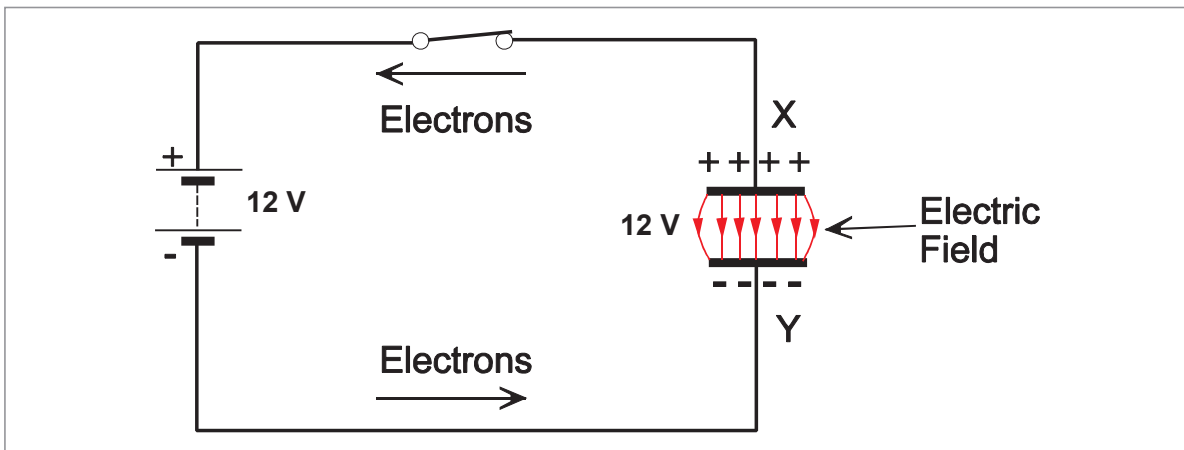


Figure 3.6

After a short time the difference in charge between the plates results in a potential difference existing between the plates. The flow of electrons will reduce and stop when the potential difference between the plates is equal to the supply voltage. The capacitor is now fully charged, current has stopped flowing, the plates are said to be charged and there exists an **electric field** between the plates. The **capacitor is now blocking DC flow**.

If the switch is opened and the capacitor is disconnected from the battery, it holds its charge: **a capacitor stores electrical energy by the formation of an electric field between the plates**. The capacitor will only discharge if it is now connected to an external circuit.

Capacitor in an AC Circuit

Figure 3.7 shows the battery replaced with an Alternating Current Supply. A light bulb is placed in series with the supply and the capacitor.

As the terminals X and Y are now changing from positive to negative at a rate depending on the frequency of the supply, current is first flowing in one direction, reversing and flowing in the opposite direction. The capacitor is charging in one direction, discharging and then charging in the opposite direction. This process continues until the supply is disconnected. The bulb will be continuously ON. **Current flows in the wires but no current flows through the dielectric.**

Therefore: **A capacitor appears to pass AC**

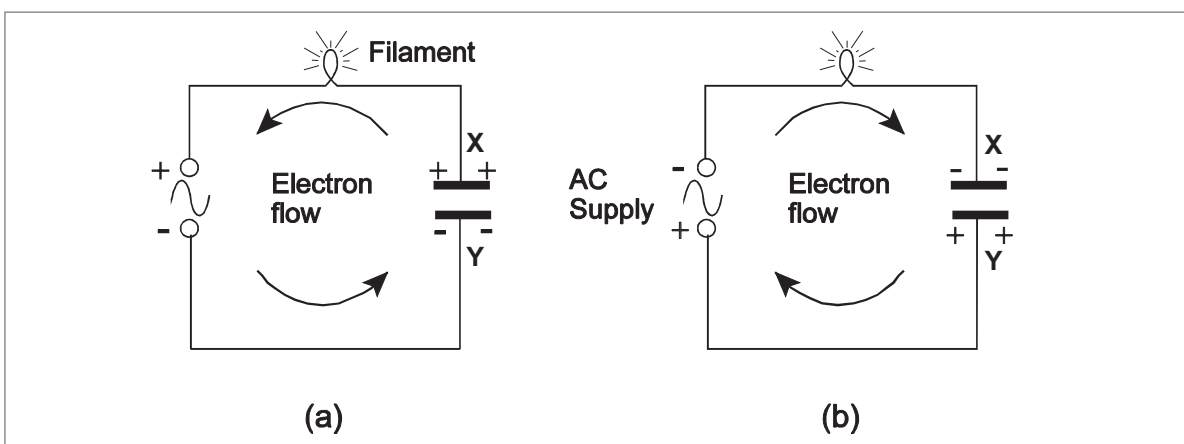


Figure 3.7

Capacitors in Parallel

Capacitors connected in parallel are effectively increasing the area of the plates. The total capacitance C_T can be found by adding the individual capacitances:

$$C_T = C_1 + C_2 \text{ etc}$$

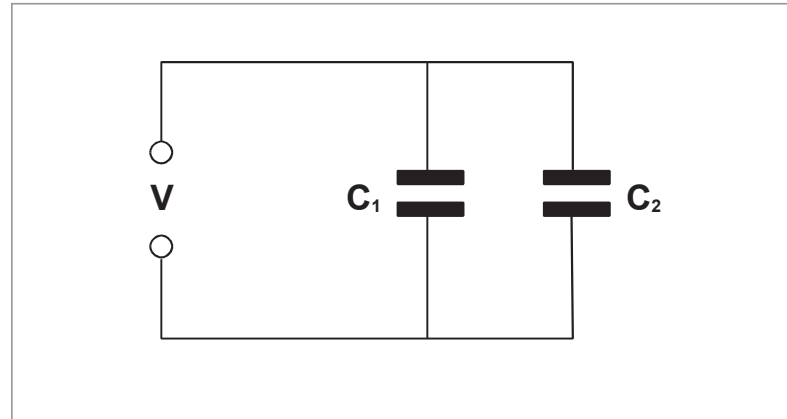


Figure 3.8

Capacitors in Series

Capacitors in series have effectively increased the distance between the plates and therefore the total capacitance has decreased. The total capacitance is found by using the formula for resistances in parallel:

$$\frac{1}{C_T} = \frac{1}{C_1} + \frac{1}{C_2} \text{ etc}$$

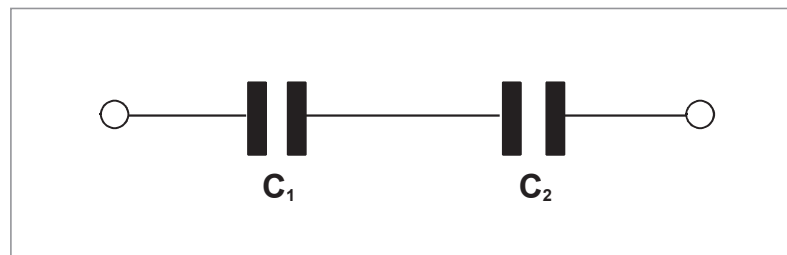


Figure 3.9